

SUPPLEMENTAL INFORMATION

Toward a precise theory of imprecise representations

Arthur Prat-Carrabin^{1,*} and Samuel J. Gershman¹

¹Department of Psychology and Center for Brain Science, Harvard University, Cambridge, MA

^{*}email: arthurpc@fas.harvard.edu

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Optimization problem: solution

The endogenous efficient-coding optimization problem is

$$\min_{n, I_1} L_{f,m,p}[nI_1] + \lambda n \quad \text{subject to} \quad \int \sqrt{I_1(x)} dx \leq \sqrt{K}, \quad (1)$$

where

$$L_{f,m,p}[nI_1] = \frac{1}{p} \int \frac{f(x)^m}{(nI_1(x))^{p/2}} dx. \quad (2)$$

Solving first for $I_1(x)$, we obtain

$$I_1(x) = K \frac{f(x)^\gamma}{Z_{f,\gamma}^2}, \quad (3)$$

where

$$\gamma = \frac{2m}{p+1} \quad (4)$$

and

$$Z_{f,\gamma} = \int f(x)^{\gamma/2} dx. \quad (5)$$

Substituting $I_1(x)$ in the optimization problem above, we obtain the problem for n :

$$\min_n \frac{Z_{f,\gamma}^{p+1}}{p K^{p/2}} \frac{1}{n^{p/2}} + \lambda n. \quad (6)$$

Solving for n , we obtain

$$n = \left(\frac{Z_{f,\gamma}^{p+1}}{2\lambda K^{p/2}} \right)^{\frac{2}{p+2}}. \quad (7)$$

Thus the total Fisher information $I(x) = nI_1(x)$ is

$$I(x) = \frac{f(x)^{\gamma_{task}}}{C_{f,m,p}}, \quad (8)$$

where

$$C_{f,m,p} = \left(\frac{2\lambda Z_{f,\gamma}}{K} \right)^{\frac{2}{p+2}}. \quad (9)$$

For a uniform prior of width w , $f(x) = 1/w$ on its support, and therefore

$$Z_{f,\gamma} = w^{1-\gamma/2}.$$

It follows that

$$n \propto w^{\frac{2(p+1-m)}{p+2}}.$$

Thus, for squared-error estimation tasks ($m = 1, p = 2$), $n \propto w$, whereas for choice tasks ($m = 2, p = 2$), $n \propto w^{1/2}$. Moreover, for a uniform prior the one-sample Fisher information satisfies $I_1 \propto w^{-2}$, so the total Fisher information scales as $I \propto w^{-1}$ and $I \propto w^{-3/2}$, respectively.

Bias and imprecision

Given the approximation to the bias (asymptotically for large n),

$$\text{bias}(x) \approx \frac{1}{I(x)} \left[\frac{f'(x)}{f(x)} - \frac{I'(x)}{I(x)} \right], \quad (10)$$

the bias with the optimal encoding (Eq. 8) is approximately

$$\text{bias}_\gamma(x) \approx (1 - \gamma) C_{f,m,p} \frac{f'(x)}{f(x)^{\gamma+1}}. \quad (11)$$

Equivalently,

$$\text{bias}_\gamma(x) \approx C_{f,m,p} \frac{\gamma - 1}{\gamma} \frac{d}{dx} f(x)^{-\gamma}. \quad (12)$$

With a task characterized by m' , p' , and $\gamma = \frac{2m'}{p'+1}$, we have

$$\sigma_{\gamma'}^2(x) = \frac{1}{I(x)} = C_{f,m',p'} f(x)^{-\gamma'}. \quad (13)$$

Therefore,

$$\text{bias}_\gamma(x) \approx (\gamma - 1) \mathcal{C} \frac{d}{dx} \sigma_{\gamma'}(x)^{2\gamma/\gamma'}, \quad (14)$$

where

$$\mathcal{C} = \frac{1}{\gamma} \frac{C_{f,m,p}}{C_{f,m',p'}^{\gamma/\gamma'}}. \quad (15)$$

Expected loss in discrimination tasks

We present the derivations of the expected loss functions in discrimination tasks [1]. Asymptotically for large n , the distribution of the Bayesian-mean estimator $\hat{x}_i$ of a stimulus x_i is a Gaussian distribution with mean x_i and variance $1/I(x_i)$. The difference between two estimates, $\hat{x}_1 - \hat{x}_0$, is then normally distributed around $x_1 - x_0$, with variance $1/I(x_0) + 1/I(x_1)$. Let

$$z = \frac{x_1 - x_0}{\sqrt{\frac{1}{I(x_0)} + \frac{1}{I(x_1)}}}. \quad (16)$$

The probability of mis-ordering x_1 and x_2 is $P(\text{error}|x_0, x_1) = \Phi(-|z|)$, where Φ the standard normal CDF. In choice tasks, if the observer chooses the ‘incorrect’ stimulus, then the loss is $|x_1 - x_0|$ (or some proportional quantity). In classification tasks, the loss in case of an error is a fixed penalty. Thus the expected loss is

$$\iint P(\text{error}|x_0, x_1) |x_1 - x_0|^{p-1} \pi(x_0) \pi(x_1) dx_0 dx_1, \quad (17)$$

where $p = 2$ in choice tasks and $p = 1$ in classification tasks.

Given x_0 , the loss averaged over x_1 is thus approximately

$$\int \Phi(-|z|) |z|^{p-1} \left(\frac{1}{I(x_0)} + \frac{1}{I(x_1)} \right)^{p/2} \pi(x_1) dz \quad (18)$$

$$\simeq \frac{\pi(x_0)}{I(x_0)^{p/2}} \int 2^{p/2} \Phi(-|z|) |z|^{p-1} dz \quad (19)$$

$$\propto \frac{\pi(x_0)}{I(x_0)^{p/2}}. \quad (20)$$

Averaging over x_0 we thus obtain the expected loss

$$\int \frac{\pi(x_0)}{I(x_0)^{p/2}} dx_0, \quad (21)$$

with $p = 2$ in choice tasks and $p = 1$ in classification tasks.

Visual working memory

Derivation of scaling laws

Fisher information of a population with Gaussian tuning curves

Let s be a one-dimensional stimulus. We are interested in the encoding capacity of a network of M neurons representing this stimulus. Specifically we consider neurons characterized by a preferred stimulus, $\tilde{s}$, a gain $g(\tilde{s})$, a tuning width $\sigma(\tilde{s})$, and whose mean responses are Gaussian functions, as

$$r_{\tilde{s}}(s) = g(\tilde{s}) \exp \left[-\frac{(s - \tilde{s})^2}{2\sigma(\tilde{s})^2} \right]. \quad (22)$$

With Poisson spike counts, the Fisher information contributed by one neuron is

$$I_{\tilde{s}}(s) = \frac{[\partial_s r_{\tilde{s}}(s)]^2}{r_{\tilde{s}}(s)} = r_{\tilde{s}}(s) \frac{(s - \tilde{s})^2}{\sigma(\tilde{s})^4}. \quad (23)$$

Let $d(\tilde{s})$ denote the density of neurons with preferred stimuli $\tilde{s}$. In the continuum approximation, the total Fisher information is

$$I_1(s) \simeq \int g(\tilde{s}) d(\tilde{s}) \exp \left[-\frac{(s - \tilde{s})^2}{2\sigma(\tilde{s})^2} \right] \frac{(s - \tilde{s})^2}{\sigma(\tilde{s})^4} d\tilde{s} \simeq \sqrt{2\pi} \frac{g(s) d(s)}{\sigma(s)}, \quad (24)$$

where we have assumed that $g(\tilde{s})$, $d(\tilde{s})$, and $\sigma(\tilde{s})$ vary slowly across a tuning width.

Tiling property

We posit that the network verifies a ‘tiling’ property. Near a stimulus value s , the neurons that respond appreciably are those whose preferred stimuli lie within a range of order $\sigma(s)$. Since there are $d(s)$ preferred stimuli per unit length, the effective number of tuning curves overlapping at s is proportional to $d(s)\sigma(s)$. The tiling property is the requirement that this overlap be the same everywhere, i.e.,

$$d(s)\sigma(s) = \tau. \quad (25)$$

In other words, the width of tuning curves is inversely proportional to their density. With lower density (i.e., tuning curves that are ‘further’ apart), the width is larger, and conversely. Consequently,

$$I_1(s) = \frac{\sqrt{2\pi} g(s)}{\tau} d(s)^2. \quad (26)$$

We now assume that d and g are constant. For a uniform stimulus space of width w tiled by M neurons, $d = M/w$, and hence the population Fisher information is

$$I_1 = \frac{\sqrt{2\pi} g}{\tau} \left(\frac{M}{w} \right)^2. \quad (27)$$

We highlight the quadratic dependence of the Fisher information on the number of neurons, M , which follows from the tiling property. This implies that dividing the M neurons into two independently re-tiled subpopulations of $M/2$ neurons gives each subpopulation Fisher information $I_1/4$ (rather than $I_1/2$ as would be the case under a linear allocation of Fisher information).

Estimation task with N variables

We now consider the case of encoding N one-dimensional variables. A fraction ρ_i of the population of M neurons is dedicated to variable i (thus $\sum_{i=1}^N \rho_i = 1$). Assuming that each subpopulation verifies the tiling property, its Fisher information is thus

$$I_1^{(i)} = I_1 \rho_i^2. \quad (28)$$

With n independent population samples,

$$I^{(i)} = n I_1 \rho_i^2. \quad (29)$$

We consider an estimation task with L_p loss, in which one of the N variables is randomly probed (with equal probabilities). With Fisher information I , the expected loss is proportional to $I^{-p/2}$. Thus the observer chooses the number of samples, n , and the allocation $\{\rho_i\}_{i=1}^N$, so as to solve the optimization problem

$$\min_{\substack{n > 0, \rho_i > 0 \\ \sum_i \rho_i = 1}} \frac{2}{p} \sum_{i=1}^N \frac{1}{N(n I_1 \rho_i^2)^{p/2}} + \lambda n M. \quad (30)$$

Given n , the optimal allocation is uniform, i.e., $\rho_i = 1/N$, and thus $I_1^{(i)} = I_1/N^2$. The remaining problem for n is

$$\min_{n>0} \frac{2}{p} \left(\frac{nI_1}{N^2} \right)^{-p/2} + \lambda nM. \quad (31)$$

Its first-order condition gives

$$n = \left[\frac{I_1^{-p/2} N^p}{\lambda M} \right]^{2/(p+2)}. \quad (32)$$

Therefore the optimized Fisher information for each variable is

$$I^{(i)} = \left(\frac{I_1}{\lambda M} \right)^{2/(p+2)} N^{-4/(p+2)}. \quad (33)$$

The Fisher information is thus proportional to $N^{-2\alpha}$, where $\alpha = \frac{2}{p+2}$, and thus $\alpha \leq 1$. We note that without the tiling property, $I_1^{(i)}$ would be proportional to ρ_i instead of ρ_i^2 , and as a result the Fisher information would be instead proportional to $N^{-2\tilde{\alpha}}$, where $\tilde{\alpha} = \frac{1}{p+2}$, and thus $\tilde{\alpha} \leq 1/2$. Hence without the tiling property, this model cannot explain exponents greater than $1/2$. It has been noted in the literature on visual working memory, however, that when fitting to empirical data models of this kind, very often the exponent is greater than $1/2$, and less than 1 [2–6]. The model with the tiling property is consistent with these observations, hence it is the one we consider here.

Change-localization task

We now consider a task in which the participant is presented sequentially with two displays of N items with various orientations. Between the presentations, exactly one item changes, by an amount Δ , and the observer is asked to report which item has changed. The per-display, one-sample Fisher information for each item is I_1/N^2 . If the two displays have independent noise, the Fisher information for their difference is therefore

$$I_{1,N} = \left(\frac{N^2}{I_1} + \frac{N^2}{I_1} \right)^{-1} = \frac{I_1}{2N^2}. \quad (34)$$

With n samples, the total Fisher information for an item is $I = nI_{1,N}$. Let δ_i be the perceived change for item i , and suppose item N is the changed item. Then, approximately,

$$\delta_N \sim \mathcal{N}(\Delta, I^{-1}), \quad \text{and} \quad \delta_i \sim \mathcal{N}(0, I^{-1}), \quad i = 1, \dots, N-1. \quad (35)$$

The observer selects the item with the largest absolute measured change, and thus misidentifies the item if $|\delta_N|$ is smaller than one of the other absolute measured changes. The probability of an error is thus

$$\Pr(\text{error} \mid \Delta) = \Pr \left(|\delta_N| < \max_{1 \leq i \leq N-1} |\delta_i| \mid \Delta \right) \quad (36)$$

$$= \int_{-\infty}^{\infty} p(\delta_N \mid \Delta) \left[1 - \prod_{i=1}^{N-1} \Pr(|\delta_i| < |\delta_N|) \right] d\delta_N \quad (37)$$

$$= \int_{-\infty}^{\infty} p(\delta_N \mid \Delta) \left(1 - \left[2\Phi(|\delta_N|\sqrt{I}) - 1 \right]^{N-1} \right) d\delta_N. \quad (38)$$

Equivalently,

$$\Pr(\text{error} \mid \Delta) = \int_0^{\infty} [\phi(t - \Delta\sqrt{I}) + \phi(t + \Delta\sqrt{I})] (1 - [2\Phi(t) - 1]^{N-1}) dt, \quad (39)$$

where ϕ and Φ are the standard-normal density and distribution function.

If the change Δ is uniform on $[-w/2, w/2]$, then the loss is

$$L(I, N) = \frac{2}{w} \int_0^{w/2} \Pr(\text{error} \mid \Delta) d\Delta. \quad (40)$$

In the small-noise regime, errors occur only near $\Delta = 0$, so the upper limit can be extended to infinity. Using

$$\int_0^{\infty} [\phi(t - \Delta\sqrt{I}) + \phi(t + \Delta\sqrt{I})] d\Delta = \frac{1}{\sqrt{I}}, \quad (41)$$

we have

$$L(I, N) \simeq \frac{2}{w\sqrt{I}} \int_0^\infty \{1 - [2\Phi(t) - 1]^{N-1}\} dt. \quad (42)$$

Let

$$a_N \equiv \int_0^\infty (1 - [2\Phi(t) - 1]^{N-1}) dt. \quad (43)$$

Then

$$L(I, N) \simeq \frac{2a_N}{w\sqrt{I}}. \quad (44)$$

We note that for independent, standard normal random variables $Z_j \sim \mathcal{N}(0, 1)$,

$$\Pr\left(\max_{1 \leq j \leq N-1} |Z_j| \leq t\right) = [2\Phi(t) - 1]^{N-1}. \quad (45)$$

The tail-sum formula for non-negative random variables ($\mathbb{E}[X] = \int_0^\infty \Pr(X > t) dt$) shows that

$$a_N = \mathbb{E}\left[\max_{1 \leq j \leq N-1} |Z_j|\right] = \int_0^\infty (1 - [2\Phi(t) - 1]^{N-1}) dt. \quad (46)$$

Thus a_N is the expected largest standardized measured change, among the $N - 1$ unchanged items. We note that for large N , we have

$$a_N \sim \sqrt{2 \log(2N)}. \quad (47)$$

This does not, however, provide an accurate approximation for the numbers of items we consider here ($N \leq 8$).

Since $I = nI_{1,N}$, the efficient-coding problem is

$$\min_{n>0} \frac{2a_N}{w\sqrt{nI_{1,N}}} + \lambda nM. \quad (48)$$

We obtain

$$n = \left(\frac{a_N}{w\lambda M \sqrt{I_{1,N}}} \right)^{2/3}, \quad (49)$$

and hence

$$I = nI_{1,N} = \left(\frac{a_N I_{1,N}}{w\lambda M} \right)^{2/3} = \left(\frac{a_N I_1}{2w\lambda M} \right)^{2/3} N^{-4/3}, \quad (50)$$

i.e.,

$$I \propto N^{-4/3} a_N^{2/3}. \quad (51)$$

Without the tiling property, $I_{1,N} \propto N^{-1}$ and $I \propto N^{-2/3} a_N^{2/3}$. In the main text, we denote $\alpha_N \equiv a_N^{2/3}$.

Models

We now detail the variable-precision (VP) models that we fit and simulate, and which follow the literature [3, 4, 7]. For estimation tasks, the mean Fisher information is modeled as

$$I = I_1 N^{-2\alpha}, \quad (52)$$

consistent with the derivations above (with $\alpha = \frac{2}{p+2}$). We posit trial-to-trial variability in the Fisher information, and denote by τ_{VP} the parameter governing the scale of this variability. Given the Fisher information I_{trial} on one trial, the distribution of the error, ε , is von Mises, as

$$p(\varepsilon | I_{\text{trial}}) = \frac{\exp(\kappa \cos \varepsilon)}{2\pi I_0(\kappa)}, \quad \text{where} \quad I_{\text{trial}} = \kappa \frac{I_1(\kappa)}{I_0(\kappa)}. \quad (53)$$

The concentration parameter κ is obtained by numerically inverting the second relation. We consider different variable-precision models, each characterized by the distribution of I_{trial} around I (see below). The error density is the mixture

$$p(\varepsilon | I, \tau_{VP}) = \int_0^\infty p(\varepsilon | I_{\text{trial}}) p(I_{\text{trial}} | I, \tau_{VP}) dI_{\text{trial}}, \quad (54)$$

with the integral replaced by a sum for a discrete distribution.

VP-Gamma

This model assumes a Gamma distribution, as

$$I_{\text{trial}} \sim \text{Gamma}\left(\frac{I}{\tau_{VP}}, \tau_{VP}\right), \quad (55)$$

which implies

$$\mathbb{E}[I_{\text{trial}}] = I, \quad \text{Var}(I_{\text{trial}}) = I\tau_{VP}. \quad (56)$$

VP-Poisson

Here the trial Fisher information takes the values $I_{\text{trial}} = \tau_{VP}k$, with

$$\Pr(k) = \text{Poisson}\left(k; \frac{I}{\tau_{VP}}\right), \quad k = 0, 1, \dots \quad (57)$$

It has the same mean and variance as VP-Gamma. Its predicted error density is

$$p(\varepsilon \mid I, \tau_{VP}) = \sum_{k=0}^{\infty} \text{Poisson}\left(k; \frac{I}{\tau_{VP}}\right) p(\varepsilon \mid I_{\text{trial}} = \tau_{VP}k). \quad (58)$$

In particular, $k = 0$ gives zero Fisher information and therefore a uniform error distribution.

VP-negative binomial

This model interpolates between VP-Gamma and VP-Poisson. With

$$r = \frac{I}{\tau_{VP}(1 - p_{\text{NB}})}, \quad (59)$$

the trial Fisher information takes the values $I_{\text{trial}} = \tau_{VP}p_{\text{NB}}k$, weighted by

$$\Pr(k) = \frac{\Gamma(k+r)}{\Gamma(r)k!} p_{\text{NB}}^r (1 - p_{\text{NB}})^k. \quad (60)$$

Again, its mean is I and its variance is $I\tau_{VP}$. It approaches VP-Gamma as $p_{\text{NB}} \rightarrow 0$ and VP-Poisson as $p_{\text{NB}} \rightarrow 1$.

Motor noise

We also consider models in which there is motor noise in addition to the encoding noise. In these models, an independent von Mises response error with concentration κ_r is added to the error. Conditional on I_{trial} , the resulting density is

$$p(\varepsilon \mid I_{\text{trial}}, \kappa_r) = \frac{I_0\left(\sqrt{\kappa^2 + \kappa_r^2 + 2\kappa\kappa_r \cos \varepsilon}\right)}{2\pi I_0(\kappa)I_0(\kappa_r)}. \quad (61)$$

This density replaces the conditional von Mises density in the mixtures above.

Apart from the exponent α , which we argue is determined by the task, the parameters of these models are I_1 and τ_{VP} ; models with motor-noise add κ_r , and the negative-binomial model adds p_{NB} .

Model fitting

We fit these models to the orientation data obtained by van den Berg *et al.*, 2012 [3] and by Bays, 2014 [5]. In both experiments, we find that the best-fitting model (as measured by the BIC) is the VP-Poisson one (with motor noise for the data from Ref. [3], and without motor noise for the data from Ref. [5]).

Allowing α to be a free parameter, we find that its best-fitting estimate is 0.680 for the data from Ref. [3]. This value is close to the prediction $\alpha = 2/3$ which follows from the score function specified in this experiment (a linear function of the absolute error, i.e., $p = 1$), and from the relation we derive, i.e., $\alpha = \frac{2}{p+2}$. In fact, fixing α to this predicted value yields a lower BIC by 9.21.

As for the data from Ref. [5], the best-fitting value of α is 0.487, which is close to the prediction derived from assuming a quadratic loss, i.e., $p = 2$ and thus $\alpha = 1/2$. Fixing α to $1/2$ lowers the BIC by 7.96.

Together, these results suggest that the different scaling exponents in the two experiments can be explained by different task losses, within the same VP-Poisson encoding model. These exponents are quantitatively predicted from these losses, via the efficient-coding derivations presented above.

VP-Poisson model for change localization

We use the same VP-Poisson encoding model for the change-localization experiment [3]. The difference from the estimation tasks lies in the two successive displays and in the decision rule. The derivation above predicts the following mean effective comparison information at set size N :

$$I = I_2 \left(\frac{N}{2} \right)^{-4/3} \left(\frac{a_N}{a_2} \right)^{2/3}, \quad (62)$$

where I_2 is the mean comparison information at $N = 2$.

For each item i , we draw

$$k_i \sim \text{Poisson} \left(\frac{2I}{\tau_{VP}} \right), \quad (63)$$

and set the per-display Fisher information to $\tau_{VP} k_i$ (the same precision draw is used for the two displays, but their measurement errors are independent). The Fisher information for their difference is therefore $\tau_{VP} k_i / 2$, whose mean is I . The corresponding von Mises concentration κ_i is obtained from the Fisher-information relation given above.

Let T denote the item that changes, let Δ denote its change, and let δ_i denote the measured change for item i . Conditional on $T = i$ and Δ , the density of δ_i is the circular convolution of the two independent von Mises measurement distributions:

$$p(\delta_i | T = i, \Delta) = \int_{-\pi}^{\pi} \frac{e^{\kappa_i \cos x}}{2\pi I_0(\kappa_i)} \frac{e^{\kappa_i \cos(x+\delta_i-\Delta)}}{2\pi I_0(\kappa_i)} dx = \frac{I_0(\kappa_i \sqrt{2 + 2 \cos(\delta_i - \Delta)})}{2\pi I_0(\kappa_i)^2}. \quad (64)$$

If $T \neq i$, item i does not change, so the same expression with $\Delta = 0$ gives

$$p(\delta_i | T \neq i) = \frac{I_0(\kappa_i \sqrt{2 + 2 \cos \delta_i})}{2\pi I_0(\kappa_i)^2}. \quad (65)$$

The true change Δ is uniform on the circle. Marginalizing over it gives

$$p(\delta_i | T = i) = \frac{1}{2\pi} \int_{-\pi}^{\pi} p(\delta_i | T = i, \Delta) d\Delta = \frac{1}{2\pi}. \quad (66)$$

Conditional on $T = i$, the measurements are independent across items. Hence

$$p(\{\delta_j\} | T = i) = \frac{1}{2\pi} \prod_{j \neq i} p(\delta_j | T \neq j). \quad (67)$$

All items have the same prior probability of being the changed item, so Bayes' rule gives

$$p(T = i | \{\delta_j\}) \propto \prod_{j \neq i} p(\delta_j | T \neq j). \quad (68)$$

The product can be written as

$$\prod_{j \neq i} p(\delta_j | T \neq j) = \frac{\prod_{j=1}^N p(\delta_j | T \neq j)}{p(\delta_i | T \neq i)}. \quad (69)$$

The product in the numerator is the same for every candidate i . Therefore

$$p(T = i | \{\delta_j\}) \propto \frac{1}{p(\delta_i | T \neq i)}, \quad (70)$$

and the Bayesian observer reports

$$\hat{T} = \underset{i}{\operatorname{argmax}} \frac{I_0(\kappa_i)^2}{I_0(\kappa_i \sqrt{2 + 2 \cos \delta_i})}. \quad (71)$$

These derivations are consistent with [3]. (Here we removed from notations the conditioning on the realized $\{\kappa_i\}$.) We evaluate the proportion of correct responses by Monte Carlo simulation, averaging over both the k_i and the measurement noise. We fit the parameters I_2 and τ_{VP} to the proportions of correct responses shown in Fig. 6B of Ref. [3]. For each parameter pair, we estimated the predicted proportion at every N and Δ using 200,000 simulations, and minimized the squared residuals divided by the squared SEMs.

Best-fitting scalings

For both the estimation task and the change-localization task, one difference with the model examined in Ref. [3] is the dependence of the Fisher information on N . They posit a power-law scaling, with an exponent left as a free parameter. In the estimation task they find a best-fitting exponent of 1.41 ± 0.15 (mean $\pm$ sem), corresponding to $\alpha = 0.705 \pm 0.075$ with our convention (their convention is $I \propto N^{-\alpha}$ rather than $I \propto N^{-2\alpha}$). This is close to the value we obtain, and to our prediction $\alpha = 2/3$.

In the change-localization task we predict a scaling in $N^{-4/3} a_N^{2/3}$. Approximating this relation with a power-law, $N^{-2\alpha}$, over $N = 2, 4, 6, 8$, we find a value of $\alpha = 0.479$. Ref. [3] report an exponent of 0.993 ± 0.075 , i.e., $\alpha = 0.497 \pm 0.038$, very close to our prediction.

Shepard's universal law of generalization

Hypothesis

Given a generalization matrix G , suppose there exist points $z_1, \dots, z_n \in \mathbb{R}^d$ such that

$$G_{ij} = \exp(-\|z_i - z_j\|^2). \quad (72)$$

Define

$$L_{ij} = -\log G_{ij} = \|z_i - z_j\|^2. \quad (73)$$

The question is whether such points exist, i.e. whether L is a matrix of squared Euclidean distances, and if so how to recover them. Without loss of generality we assume that the points are centered, i.e.,

$$\sum_{i=1}^n z_i = 0. \quad (74)$$

Gram matrix

Expanding the square,

$$L_{ij} = \|z_i\|^2 + \|z_j\|^2 - 2\langle z_i, z_j \rangle. \quad (75)$$

Let Z be the $n \times d$ matrix whose i -th row is $z_i^\top$, and let

$$B = ZZ^\top \in \mathbb{R}^{n \times n}, \quad B_{ij} = \langle z_i, z_j \rangle.$$

This is the Gram matrix of the configuration. A Gram matrix is positive semidefinite (PSD; as $a^\top Ba = \|Z^\top a\|^2 \geq 0$ for all $a \in \mathbb{R}^n$). Conversely, every PSD matrix is the Gram matrix of some set of points. So testing whether L is squared-Euclidean is equivalent to testing whether the corresponding B is PSD. Specifically, if B has a negative eigenvalue, it is not PSD, thus it is not a Gram matrix, so no exact squared-Euclidean representation of L exists. If it is PSD, then we can write the eigendecomposition $B = U\Lambda U^\top$ and set $\tilde{Z} = U\Lambda^{1/2}$. Then $\tilde{Z}\tilde{Z}^\top = B$, and the rows of $\tilde{Z}$ are explicit coordinates whose squared distances reproduce L . We now derive B from L .

Recovering B from L

In terms of B ,

$$L_{ij} = B_{ii} + B_{jj} - 2B_{ij}, \quad \text{hence} \quad 2B_{ij} = B_{ii} + B_{jj} - L_{ij}. \quad (76)$$

We know L , so the only missing ingredient is the diagonal, $B_{ii} = \|z_i\|^2$, i.e., the squared distance of each point from the centroid. It can be recovered from L alone. We denote the row means and grand mean of L by

$$\bar{L}_i = \frac{1}{n} \sum_j L_{ij}, \quad \bar{L} = \frac{1}{n^2} \sum_{i,j} L_{ij}. \quad (77)$$

Averaging L_{ij} over j ,

$$\bar{L}_i = \|z_i\|^2 + \frac{1}{n} \sum_j \|z_j\|^2 - \frac{2}{n} \left\langle z_i, \sum_j z_j \right\rangle = \|z_i\|^2 + \frac{1}{n} \sum_j \|z_j\|^2, \quad (78)$$

the last term vanishing because the points are centered. Averaging this equality over i ,

$$\bar{L} = \frac{1}{n} \sum_i \bar{L}_i = \frac{2}{n} \sum_i \|z_i\|^2, \quad \text{i.e.} \quad \frac{1}{n} \sum_j \|z_j\|^2 = \frac{\bar{L}}{2}. \quad (79)$$

Therefore

$$\bar{L}_i = \|z_i\|^2 + \frac{\bar{L}}{2}, \quad \text{so} \quad B_{ii} = \|z_i\|^2 = \bar{L}_i - \frac{\bar{L}}{2}. \quad (80)$$

Thus we obtain

$$B_{ij} = -\frac{1}{2} (L_{ij} - \bar{L}_i - \bar{L}_j + \bar{L}). \quad (81)$$

This can be written in matrix form. Let $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$, where $\mathbf{1} \in \mathbb{R}^n$ is the vector of ones, so that $\mathbf{1}\mathbf{1}^\top$ is the all-ones matrix. (H is a centering matrix: for any $v \in \mathbb{R}^n$, $(Hv)_i = v_i - \bar{v}$. It satisfies $H^\top = H$, $H^2 = H$, and $H\mathbf{1} = 0$.) We have:

$$B = -\frac{1}{2} H L H. \quad (82)$$

Empirical data

For each of the nine experiments, we compute the generalization matrix G using the reported confusion matrix. (To estimate the conditional probabilities p_{ij} , we add a pseudocount of 1 to the raw counts, equivalent to Bayesian estimation of p_{ij} under a Dirichlet-Multinomial model.) Given G , we compute B and examine its spectrum. We find that seven of the nine datasets have one or two negative eigenvalues, while two have none. The negative eigenvalues are generally small relative to the positive spectrum. To measure this, we compute the fraction of negative spectral mass, as $\sum_{\lambda_k < 0} |\lambda_k| / \sum_k |\lambda_k|$, where λ_k denote the eigenvalues. Its median is 0.9% and its mean is 1.5% (SD: 2.0%). We conclude that the violations to the squared-distance hypothesis are small, and probably originate in sampling noise.

Thus we only keep the positive eigenvalues. Specifically, let

$$\Lambda_+ = \text{diag}(\max(\lambda_1, 0), \dots, \max(\lambda_n, 0)), \quad \text{and} \quad B_+ = U \Lambda_+ U^\top. \quad (83)$$

The matrix B_+ is a positive-semidefinite approximation to B . We then obtain the corresponding coordinates as $\hat{Z} = U \Lambda_+^{1/2}$. The resulting recovered distances are $d_{ij} = \|\hat{z}_i - \hat{z}_j\|$. In the main text we plot the empirical values G_{ij} against the recovered distances d_{ij} , together with the predicted curve $G_{ij} = \exp(-d_{ij}^2)$.

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